

Fiona Mackenzie

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SKILLS

SolidWorks: CAD & 3D Modeling, Equipment Design, FEA and Simulation.

- *Applied in torsional stiffness improvements on Capstone 4x rowing hulls project.*

AutoCAD: 3D modeling and 2D technical drawings.

- Used at Dubois Chemicals Mechanical Engineering Internship.

Equipment Testing: Nozzle (Pressure vs. Flowrate) Testing, Pump Testing, Equipment Leak Testing.

- Performed during Dubois Chemicals Mechanical Engineering Internship.

Ansys Fluent: Computational Fluid Dynamics (CFD). Flow domain design, meshing and simulation.

- Applied in *Leaning Tower of Pisa Wind Load Analysis* Project.

MATLAB – Skilled in numerical Analysis and automating engineering calculations.

EDUCATION

Sept 2021 - June 2025

Western University, London Ontario - **Bachelor of Science in Mechanical Engineering**

- Relevant Coursework: Fluid Mechanics, Thermodynamics, Heat Transfer, Materials, Mechanical design, Engineering Mathematics, Finite Element Analysis (FEA), Electrical Engineering and Reverse Engineering.

WORK EXPERIENCE

DuBois Chemicals - Engineering Internship, Project Management Team

September 2025 - August 2026

- Completed testing and selection of nozzles for dust suppression equipment. Including flowrate and pressure calculations to determine quantity and capacity for each application point.
- Designed and developed the 3D models and engineering drawings for dust suppression application equipment including: Custom nozzle mounting plates, standardized metering system packages, hydraulic distribution manifolds, Stack Reclaimer dust suppression.
- Designed a portal stacker reclaimer dust suppression system, including nozzles mounted on the booms of the machine and in the chutes of the machine. This design was coordinated with Koch Engineering, Wison Engineering, Abu Dhabi National Oil Company and DuBois Chemicals.

Linde Canada Inc. - Engineering Co-op Student, Customer Service Department

May 2022 – August 2022

- Reviewed and updated cryogenic equipment asset information in the Enterprise Asset Management system (EAM).
- Reviewed and verified Canadian Registration Numbers (CRN) for pressure retaining components listed on Linde standard drawings; obtaining copies of applicable statutory declarations from vendors as required.
- Identified gaps/opportunities between the existing Linde standard maintenance program for cryogenic systems and industry best practice.
- All tank information organized using Microsoft Excel.

Urban Life Solutions - General Labourer

May 2024 - August 2024

- Landscaping, property planning and design.
- Environmental and safety implications.
- Operating motorized tools (Trimmer, Blower, Mowers, etc.).

PROJECTS

Torsional Stiffness Improvements on 4x Rowing Hulls

- Redesigning structural elements in SolidWorks to improve stiffness-to-weight ratios.
- Conducting FEA simulations for finding deformation under torsional and lateral loads.
- Performing physical vibration testing to determine weak points along the hull.
- Running three-point bending tests on various composite materials to evaluate mechanical properties and suitability for marine applications.

Leaning Tower of Pisa - Advanced CAE Computational Heat and Fluid Flow

- Analysing air flow around the Pisa Tower, using Ansys Fluent simulations.
- Setting up geometry of domain, determining mesh properties and size, and performing simulations.

Green Hydrogen for Transportation - Research Project

- Exploring the future of green hydrogen vehicles (hydrogen fuel cells).
- Researched green hydrogen as a sustainable alternative fuel for transportation.
- Analyzed zero-emission production via renewable-powered electrolysis.
- Evaluated challenges in onboard storage and vehicle integration.

CERTIFICATIONS

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| - SolidWorks Additive Manufacturing Associate | 2024 |
| - SolidWorks Surfacing Professional | 2024 |
| - SolidWorks Simulation Associate | 2024 |
| - SolidWorks CAD Design Associate (CSWA) | 2022 |
| - CPR Certificate | 2023 |